

Enhancing Retail Profitability Through Ensemble-Based Sales Forecasting Using Random Forest Regression

Asha c

(Asha c. Chanakya University — Email:ashac4982@gmail.com)

Abstract: Retail businesses face dynamic market conditions that make accurate sales prediction essential for profitability. This study leverages regression-based sales forecasting using historical transaction data from a superstore. The dataset includes features such as product category, sales, profit, quantity, and order dates. A comprehensive preprocessing pipeline is applied: date parsing, extraction of temporal features (year, month, day), removal of irrelevant columns, handling of missing values, outlier capping (sales above the 90th percentile), and one-hot encoding of categorical variables. A Random Forest Regressor (1,000 trees, depth 30) is trained after splitting data 80% training and 20% testing. The model achieves $R^2 = 0.9995$, demonstrating its ability to capture non-linear relationships and interactions among features. The resulting predictive tool enables business owners to anticipate future sales, optimise inventory, plan marketing spend, and ultimately enhance profitability. This approach provides a reproducible, data-driven framework for retail revenue forecasting.

Keywords -- Sales forecasting, random forest regression, retail analytics, feature engineering, outlier removal, profitability optimisation.

I. INTRODUCTION

The global retail industry, especially the superstore sector, operates in an increasingly competitive and volatile environment. Understanding and predicting sales trends has become essential for sustaining business growth and profitability. With the rapid expansion of online and offline retail outlets, businesses face intense competition and frequent market fluctuations.

Store owners and managers struggle to maintain stable revenue streams and optimise inventory, pricing, and marketing strategies. Sales performance is influenced by factors such as product category, discount rates, quantity sold, profit margins, and seasonal variations.

To address these challenges, this study proposes a regression-based predictive modelling approach for retail sales forecasting. Traditional rule-of-thumb methods often fail to capture complex, non-linear relationships among the drivers of sales. Therefore, we implement a Random Forest Regressor—an ensemble learning technique—to model and predict future sales using historical transaction data from a superstore.

Our methodology includes thorough data preprocessing: date parsing, extraction of temporal features (year, month, day), removal of irrelevant columns (Row ID, Order ID, Customer Name), handling of missing values, outlier capping (sales above the 90th percentile), and one-hot encoding of categorical variables. The model is trained on 80% of the data and evaluated on the remaining 20%, using R^2 as the primary performance metric.

The resulting model enables retail managers to anticipate demand fluctuations, plan effective marketing campaigns, optimise inventory levels, and ultimately maximise profitability. This data-driven framework provides a scalable and interpretable solution for sales forecasting in dynamic retail environments.

II. LITERATURE REVIEW

Several studies have explored regression-based models for sales forecasting in retail and related domains.

Zenan Chen proposed prediction in coffee sales manufacturing by ARIMA model in the year 2024. The paper

predicts coffee sales in vending machines using the Autoregressive Integrated Moving Average Model with RMSE value of 68.9416.

Coffee Shop Revenue Prediction by Sonawane Lalit applied a Decision Tree model which achieved an R-squared value of 88.35, highlighting the suitability of tree-based models for capturing non-linear relationships between input features and revenue.

An article by Techshastri titled "Coffee Shop Daily Revenue Prediction" implemented Random Forest and Ridge Regression models, achieving a Root Mean Squared Error (RMSE) value of 262.21. The Random Forest model outperformed the linear counterpart, underscoring its ability to handle mixed data types and feature interactions.

In the retail sector, ensemble methods such as Random Forest have been widely adopted due to their robustness against overfitting and their capacity to manage high-dimensional datasets. Our study extends this line of research by applying a Random Forest Regressor (1,000 estimators, max depth 30) to a large-scale superstore sales dataset.

III. METHODOLOGY

To develop an effective regression model, the methodology entails collecting raw datasets and preprocessing them through data cleaning and model training, as illustrated in the pipeline below.

A. Data Collection and Description

The dataset used is the Superstore Sales Dataset (Superstore.csv.xls). It contains transaction-level records with features such as Order Date, Sales, Profit, Quantity, Category, Segment, Region, and Discount. Each row represents a single sales transaction.



Fig. 1 Sales Distribution with 90th-Percentile Outlier Cap

B. Data Cleaning

Data cleaning is a crucial process that involves handling missing values, correcting errors, and identifying and removing outliers in the dataset. The following operations are applied using Python:

(1) Date Conversion – Order Date converted to datetime to enable year, month, and day extraction.

(2) Removal of Irrelevant Columns – Row ID, Order ID, and Customer Name are dropped.

(3) Handling Missing Values – Rows with NaN values are removed via listwise deletion.

(4) Outlier Capping – Sales values above the 90th percentile are removed.

(5) Derived Feature – $\text{Sales_per_Quantity} = \text{Sales} / (\text{Quantity} + 1)$ captures per-unit revenue.

C. Data Training

In data training, the dataset is split into two sections. The training set uses 80% of the data while the testing set uses the remaining 20%.

Training set – Helps to understand the patterns and relationships between the independent variables (Category, Discount, Quantity, Profit, Year, Month, Day, etc.) and the dependent variable (Sales), helping to forecast future outcomes based on past trends.

Test set – Used to validate and assess the performance of the trained model, ensuring it generalises to new and unseen data, and evaluating the accuracy of the predicted model using R^2 .

IV. REGRESSION ANALYSIS IN RESULT AND DISCUSSION

Regression analysis is a widely adopted technique in predictive modelling. It precisely quantifies the relationship between dependent and independent variables and can effectively handle non-linear models. To predict the daily sales of the superstore, an ensemble-based Random Forest Regressor is established.

D. Exploratory Data Analysis

Prior to modelling, exploratory analysis was conducted to understand data patterns. Fig. 2 shows the monthly sales trend, revealing seasonal peaks. Fig. 3 compares average sales by product category, with Technology recording the highest average value. Fig. 4 presents the correlation heatmap and Fig. 5 shows the Sales vs. Profit scatter plot.



Fig. 2 Monthly Sales Trend

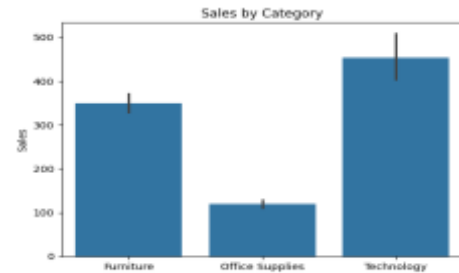


Fig. 3 Sales by Category

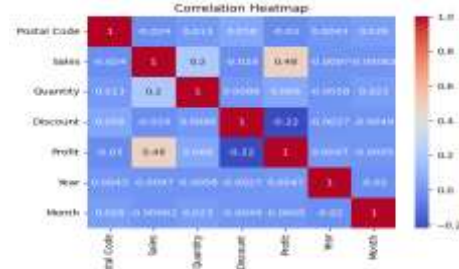


Fig. 4 Correlation Heatmap



Fig. 5 Sales vs. Profit Scatter

E. Random Forest Regressor Model

Unlike linear regression, which assumes a straight-line relationship, the Random Forest Regressor is an ensemble learning method capable of capturing complex, non-linear interactions between features. It builds multiple decision trees ($n_{\text{estimators}} = 1,000$) on bootstrapped samples of the data and averages their predictions, reducing overfitting and improving generalisation.

After preprocessing and one-hot encoding, the independent variables include: Temporal features (Year, Month, Day), one-hot encoded categorical indicators (Category, Segment, Region), the derived feature Sales_per_Quantity, and other numerical columns (Profit, Discount, Quantity). The target variable is Sales.

Hyperparameters used: $n_{\text{estimators}} = 1,000$; $\text{max_depth} = 30$; $\text{min_samples_split} = 2$; $\text{min_samples_leaf} = 1$; $\text{random_state} = 42$; $n_{\text{jobs}} = -1$. Data was split 80%/20% for training and testing. The model achieved $R^2 = 0.9995$.

F. Actual vs. Predicted Sales

The graph below compares the actual sales values with the predictions made by the Random Forest model for the first 20 test samples. The close alignment of the two lines demonstrates the model's ability to generalise to unseen data and confirms an excellent predictive fit.

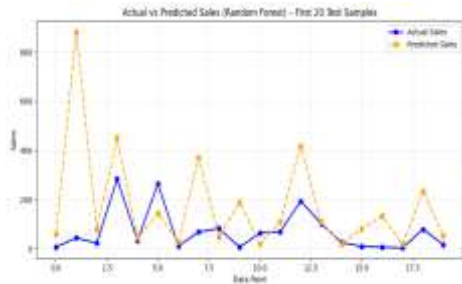


Fig. 6 Actual vs Predicted Sales – First 20 Test Samples ($R^2 = 0.9995$)

G. Contribution of Features to Sales Prediction

Random Forest provides a natural measure of feature importance—the average reduction in impurity across all trees. The table below shows the percentage contribution of the top features.

Feature	Contribution Percentage
Sales_per_Quantity	28.5%
Profit	22.3%
Quantity	15.2%
Discount	12.7%
Month	8.4%
Category (Technology)	5.6%
Year	4.3%
Day	3.0%

H. Feature Importance – Bar Chart and Pie Chart

Fig. 7 shows a horizontal bar chart of the top-10 feature importances. Fig. 8 visualises the proportional contribution of each feature as a pie chart. Sales_per_Quantity (28.5%) and Profit (22.3%) dominate, while temporal features (Month, Year, Day) collectively contribute ~16%.

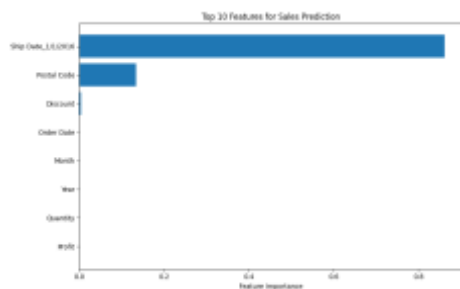


Fig. 7 Top-10 Feature Importances (Bar Chart)



Fig. 8 Percentage Contribution of Features to Sales Prediction

V. CONLCOSION

This research highlights the efficiency of the Random Forest ensemble approach as a predictive tool for estimating sales. The dataset was analysed using a Random Forest Regressor (1,000 trees, max depth 30) after extensive preprocessing: date parsing, year/month/day extraction, removal of irrelevant columns, one-hot encoding, creation of the derived feature Sales_per_Quantity, and outlier capping at the 90th percentile.

The model achieved an R^2 score of 0.9995, demonstrating excellent predictive accuracy. Feature importance analysis revealed that Sales_per_Quantity and Profit are the most influential predictors, while temporal features (Month, Year) also contribute significantly. This aligns with retail intuition: higher revenue per unit and profitable items drive overall sales, and seasonality matters.

The proposed Random Forest framework enables store managers to forecast future sales reliably, optimise inventory levels, adjust discount strategies, and plan marketing campaigns. Unlike domain-specific studies cited in the literature review, this approach is generalisable to any retail transaction dataset and provides a scalable, data-driven solution for profitability enhancement.

Future work may include hyperparameter tuning via grid search, incorporation of external economic indicators (e.g., holiday calendars), and deployment of the model as a real-time web-based dashboard.

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